NAME:
NEPTUN:

First Solution: B
Second Solution: C

1. Suppose that the price of a one-year European call option with a strike price of $100 is $15, and the price of a one-year European put option with the same strike price is $10. The one-year discount factor is 80%. Suppose that the one-year forward price in the market is $105,25. How much is the arbitrage profit future value (FV) in this case?

a) $0,8
b) $1
c) $1,25
d) $2

$F^a_1(0) = (c-p)/DF_1 + K = (15-10)/0,8 + 100 = \$106,25$
$F^m_1(0) = \$105,25$
Arbitrage FV $= F^a_1(0) - F^m_1(0) =$ **$1**

2. Suppose that one-year put options on a stock with strike prices $30 and $35 cost $4 and $7, respectively. The one-year discount factor is 100,00%. A trader take positions on these options to creat long bull spread. At which stock prices will the bull spread be profitable?

a) If the spot price at the end of the year is under $30
b) If the spot price at the end of the year is under $32
c) If the spot price at the end of the year is over $32
d) If the spot price at the end of the year is between $32 and $35

long bull spread $LP_{K1} + SP_{K2}$
$LP_{30} + SP_{35}$
CF(0) 3

		Payoff	Payoff	Profit
S(1)	LP_{30}	SP_{35}	Portfolio	Portfolio
27	3	-8	-5	-2
30	0	-5	-5	-2
32	0	-3	-3	**0**
35	0	0	0	**3**
38	0	0	0	**3**